Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/25/2006
PI Name:	Hlatky, Lynn Ph.D.		
Project Title:	Computational Modeling of Chromosome Aberrations Produced by HZE Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI affiliation name change to Steward Research and Specialty Projects Corporation (2012). PI formerly at Caritas St. Elizabeth's Medical Center, Boston, MA (2005-2011); Dana-Farber Cancer Institute, Boston, MA (prior to 2005). Subsequent name change from Steward Research and Specialty Projects Corporation to GeneSys Research Institute in 2013.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Radiation Biology and Shielding 03-OBPR-07
Start Date:	05/15/2004	**End Date:**	06/01/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hahnfeldt, Philip (Caritas St. Elizabeth's Medical Center, Tufts Univ School of Medicine) Sachs, Rainer (University of California)		
Grant/Contract No.:	NNJ06HA27G ; NNJ04HF42G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Chromosome aberrations produced by high LET radiations are directly and indirectly relevant to space exploration and risk estimation in a number of different ways. They are important as probes for DNA repair and recombination mechanisms, as biomarkers of dose and radiation quality, as factors in genomic instability and carcinogenesis, and as surrogates for cancer in risk estimation. Quantitative modeling is needed to exploit and augment the rapidly accumulating empirical data on aberration types and frequencies. This project will apply mathematical and computer algorithms to data on aberrations produced by iron ions (56Fe). The analysis will be compared to analyses of damage produced by photons, electrons, or alpha particles, with the goal of assessing signature responses to 56Fe, an important radiation encountered in space. To perform such studies, the team has tied damage endpoints to risk through our customized software and algorithms. These endpoints include: 1. aberration complexity, quantified by cycle structures; 2. apparently and truly incomplete aberrations, related to segment size distributions; 3. aberration spectra as related to chromosome geometry, to DNA double strand break (DSB) clustering depending on radiation track structure, and especially to DSB misrejoining depending on proximity; 4. mFISH color junctions for each specific chromosome pair, related to chromosome-chromosome spatial correlations in an interphase nucleus. Substantial preliminary studies have recently been carried out by our team for each aim. As remarked, we will employ the team's specialized and comprehensive software, including the highly-developed chromosome aberration simulator (CAS) software, for analyzing the aberration spectra encountered. The data to be analyzed will be for human cells in vitro, scored by mFISH or other similar techniques. A number of these data sets are being made available by many of our colleagues worldwide. This will greatly expand the range of suitable data available for analysis. We have also recently devised methods for systematically comparing mFISH data with data obtained by older methods such as FISH or solid staining and with newer methods such as color banding. For Specific Aim 1, we will replace the current approach of using "obligate" (deterministic) cycles with a more accurate computer-based approach involving cycle probabilities. For Aim 2, new methods are being developed that allow systematic comparison of apparently incomplete aberrations with corresponding complete aberrations and corresponding truly incomplete ones, even in the case of very complex aberrations. Under Aim 2 we will also consider size distributions of fragments that have one telomere and one unrejoined free end, for comparison to recent data using pan-telomeric probes; and analyze mixtures of different radiations. Aim 3 will emphasize more realistic computer algorithms for break misrejoining, bringing this aspect of aberration modeling more nearly in line with the very sophisticated models now being used by various groups for assessing the influence of track structure and chromatin geometry on DSB locations. For Aim 4 we plan to apply a method we previously used for low LET data to data for high LET, including iron ions. The method involves testing frequencies of color junction pairs for deviations from randomness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Improved biodosimetry
Task Progress:	Grant 03-OBPR-07. Computational Modeling of Chromosome Aberrations Produced by HZE Particles. Lynn Hlatky PI, Caritas St. Elizabeth's Medical Center, Tufts Univ. School of Medicine. Summary of progress reported earlier. We statistically analyzed mFISH color junction frequencies for radiation-produced chromosome aberrations as per Aim 4. For the part of Aim1 dealing with analyzing DNA damage/repair/misrepair pathways we used mFISH results on radiogenic homologue-homologue dicentrics, including results on the HZE radiation emphasized by NASA, to show that there is a dependence on homology even for mammalian cells in the G0/G1 phase of the cell cycle. We did preliminary work on Aims 2 and 3. Progress for the reporting period, 05/16/2005-05/16/2006. We continue to make progress on Aim 2 which deals with systematically comparing apparently incomplete, truly incomplete, and complete aberrations. In brief, the optimal procedure is to identify incompleteness type by looking for mismatches between centromeres and apparent telomeres involved in a particular aberration since in a complete aberration there are two telomeres of each color for each centromere of that color. Once incompleteness type has been characterized, color-by-color, adding "virtual fragments" of appropriate colors gives a fully methodical way to see what the possible completions are. An applet developed during preliminary studies, is already available on the internet at the URL http://radiobiology.math.berkeley.edu/ . It has been updated with new algorithms that give a more realistic description of the probabilities for different complete aberrations that can correspond to a given apparently incomplete mFISH aberration; previously, apparent incompleteness has been a major confounding factor in analyzing HZE data on aberrations. These results and their application to iron ion data are being reported in a paper in preparation for the Journal of Computational Biology (Levy et al. 2006). We continue to make progress on Aim 3, computer simulations of aberration formation. We have developed two geometric approaches to high-LET radiation simulation. One is based on random consecutive location of chromosomes and uses our randomly located cluster formalism; the other is based on random walk geometry for all 46 chromosomes in an interphase nucleus and uses track structure simulations. The former has recently been implemented by Java software involving about 30 classes and interfaces. This code is documented in detail at the URL http://math.berkeley.edu/ . The latter has recently been implemented by a simulated annealing approach to locating chromosomes in an interphase nucleus in which overlapping of different chromosome territories is allowed but is subjected to an energy penalty. The results are in better agreement with recent (non-radiobiological) data on chromosome territories. They are being written up in a paper for Bioinformatics (Chen et al. 2006). Another aspect of Aim 3 was DSB (DNA double-strand break) clustering at high LET and its influence on misrejoining frequencies. We have recently collaborated with A. Ponomarev and F. Cucinotta of NASA in work on how to subtract out background values of DNA fragment size distributions in PFGE experiments on radiation breakage to analyze DSB yields and clustering. The work is being prepared for IJRB (Ponomarev et al. 2006). The results also have implications for the effects of mixtures of HZE radiations and protons on DSB clustering. Cumulative bibliography. Full-Color Painting Reveals an Excess of Radiation-Induced Dicentrics Involving Homologous Chromosomes. Y. Plan, L. Hlatky, P. Hahnfeldt, R. Sachs, B. Loucas, and M. Cornforth. Int J Radiat Biol 2005; 81:613-20.

	Levy D, L Hlatky and RK Sachs (2006), Software for Cycle Analysis, http://radiobiology.berkeley.edu/ NSF SCHIP: Statistics for Chromosome Interphase Positioning. Vives S, Loucas B, Vazquez M, Brenner DJ, Sachs RK, Hlatky L, Cornforth M, Arsuaga J. Bioinformatics. 2005 21:3181-2. Submitted to J National Cancer Institute (2006). Mechanistic differences in radiation-induction of hematopoietic vs. solid cancers. Igor Shuryak, Rainer K. Sachs, Lynn Hlatky, Mark P. Little, Philip Hahnfeldt, David J. Brenner. In preparation for Bioinformatics (2006). Interphase chromosome geometry by simulated annealing. Allen Chen, Rainer Sachs, and Lynn Hlatky. In preparation for J Computational Biology (2006). D Levy, C Reeder, L Hlatky, and R Sachs Determining exchange spectra from mFISH aberration patterns. In preparation for International J Radiation Biology (2006). Subtracting Background Damage in PFGE experiments on DNA fragmentation. A.L. Ponomarev, L. Hlatky, R.K. Sachs, P. Hahnfeldt, F.A. Cucinotta.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Articles in Peer-reviewed Journals	Shuryak I, Sachs R, Hlatky L, Little M, Hahnfeldt P, Brenner D. "Mechanistic differences in radiation-induction of hematopoietic vs. solid cancers." J Natl Cancer Inst , in press, 2006. , Aug-2006